Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: DSIP (01CT1513)	**Aim: Non-Linear Noise Filtering.**	
Open-ended-Problem: 05	**Date:31-10-2026**	**Enrollment No:92400133121**

AIM: Non-Linear Noise Filtering.

Theory:

- Non-linear filtering is an image processing technique used to remove noise from digital images while preserving important image details such as edges and boundaries.
- Unlike linear filters, which calculate pixel values using weighted sums or averages of neighbouring pixels, non-linear filters modify pixel values based on the statistical or ranking properties of pixels in a local neighbourhood.
- Median Filtering and Bilateral Filtering are used to reduce noise and improve image quality.

→Median Filter←

- It replaces the central pixel value with the median value of the pixels within a selected neighbourhood (such as 3×3, 5×5, or 7×7).
- It is particularly effective in removing salt-and-pepper noise while preserving edges better than a mean filter.

→Bilateral Filter←

- It smooths an image by considering both the spatial distance between neighbouring pixels and the difference in their intensity values.
- It reduces noise while preserving edges and important image details.

- The size of the filtering window and the filter parameters influence the amount of noise removal and detail preservation.
- A smaller window generally preserves more fine details, whereas a larger window provides stronger smoothing but may remove some image details.

➢ Thus, non-linear filters are useful for reducing noise, improving image quality, and preparing images for further image processing and analysis.

Code:

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```
import cv2
import numpy as np
import matplotlib.pyplot as plt

def apply_median_filter(image, kernel_size):
    # Apply median filter to remove noise
    filtered_image = cv2.medianBlur(image, kernel_size)
    return filtered_image

def apply_bilateral_filter(image, d, sigma_color, sigma_space):
    # Apply bilateral filter to remove noise
    filtered_image = cv2.bilateralFilter(
        image, d, sigma_color, sigma_space
    )
    return filtered_image

image_paths = [
    'images/IMAGE1.png',
    'images/IMAGE2.png',
    'images/IMAGE3.png',
    'images/IMAGE5.png'
]
for index, image_path in enumerate(image_paths, start=1):

    # Load input image
    input_image = cv2.imread(image_path)

    if input_image is None:
        print("Image not found:", image_path)
        continue

    # Apply Median Filters
    median_3 = apply_median_filter(input_image, 3)
    median_5 = apply_median_filter(input_image, 5)
    median_7 = apply_median_filter(input_image, 7)

    # Apply Bilateral Filter
    bilateral = apply_bilateral_filter(
```

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```
    input_image, 9, 75, 75
)

images = [
    input_image,
    median_3,
    median_5,
    median_7,
    bilateral
]

titles = [
    'Original Image',
    'Median 3x3',
    'Median 5x5',
    'Median 7x7',
    'Bilateral Filter'
]

plt.figure(figsize=(16, 5))

for j in range(5):
    plt.subplot(1, 5, j + 1)

    rgb_image = cv2.cvtColor(
        images[j], cv2.COLOR_BGR2RGB
    )

    plt.imshow(rgb_image)
    plt.title(titles[j])
    plt.axis('off')

plt.suptitle(f'Non-Linear Filtering - Image {index}')
plt.tight_layout()
plt.show()

cv2.imwrite(f'image{index}_median3.jpg', median_3)
cv2.imwrite(f'image{index}_median5.jpg', median_5)
```

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```
    cv2.imwrite(f'image{index}_median7.jpg', median_7)
    cv2.imwrite(f'image{index}_bilateral.jpg', bilateral)

    print(f"Image {index} processed successfully.")

print("All images processed successfully!")
```

Output:

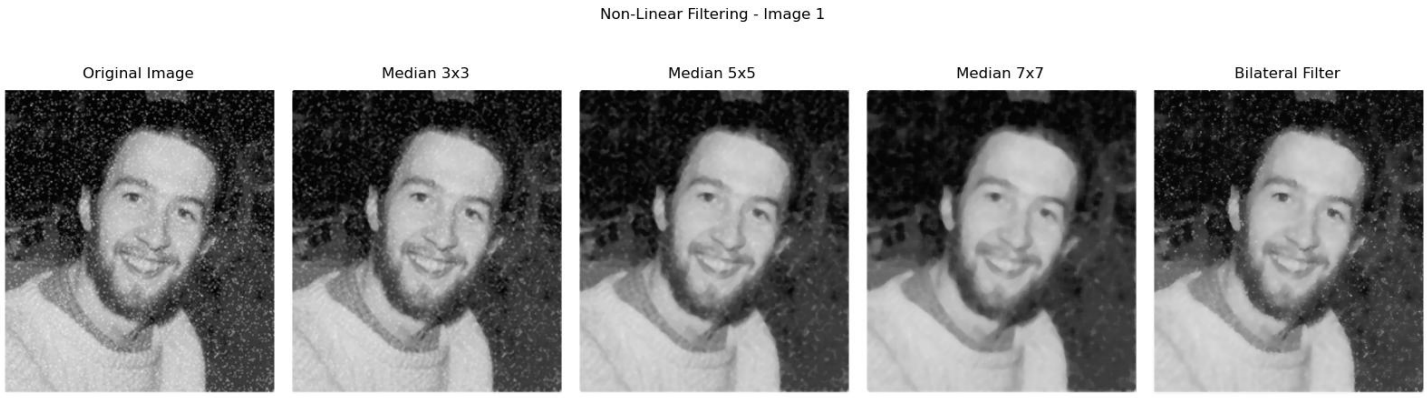


Image 1 processed successfully.

Salt-and-Pepper / Impulse Noise

Observation:

- Median 3×3 reduces noise while retaining most facial details.
- Median 5×5 provides better smoothing but slightly reduces fine details.
- Median 7×7 gives maximum smoothing and noise reduction, but causes more blurring.
- Bilateral filtering preserves facial edges better, but more noise remains compared with larger median filters.

Conclusion:

- Median 7×7 gives the strongest noise reduction but loses more fine details.
- Bilateral filtering preserves edges better while providing moderate noise reduction.

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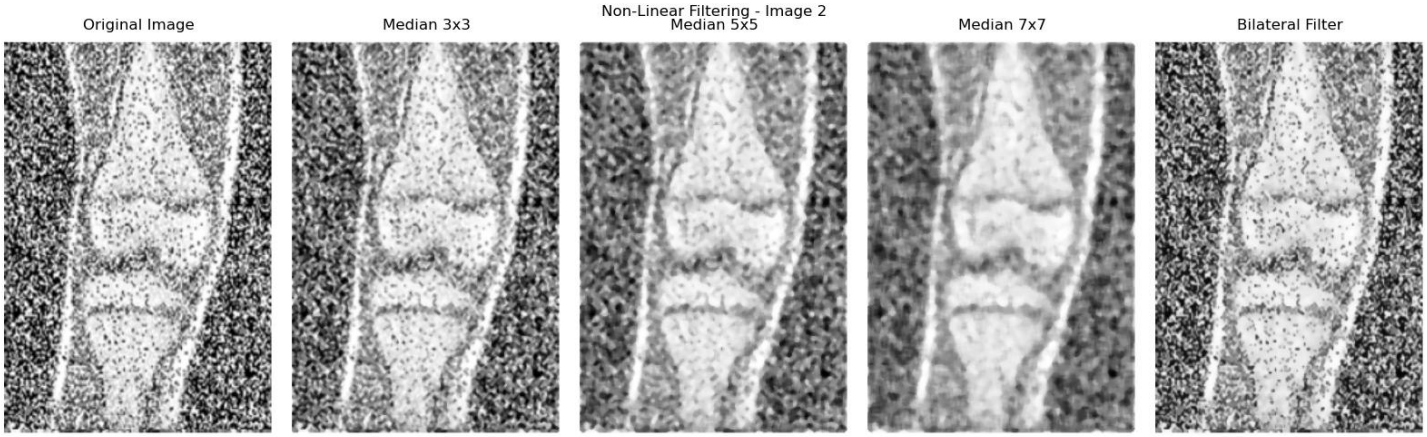


Image 2 processed successfully.

Strong Impulse / Salt-and-Pepper Noise

Observation:

- ❖ Median 3×3 provides moderate noise reduction while preserving the main structure.
- ❖ Median 5×5 gives better smoothing and makes the main structure clearer.
- ❖ Median 7×7 removes more noise but softens smaller details and boundaries.
- ❖ Bilateral filtering preserves the main edges but leaves more granular noise.

Conclusion:

- ➢ Larger median filters provide better noise reduction for this image, with Median 7×7 giving the smoothest result.
- ➢ Bilateral filtering preserves boundaries better but removes less noise.

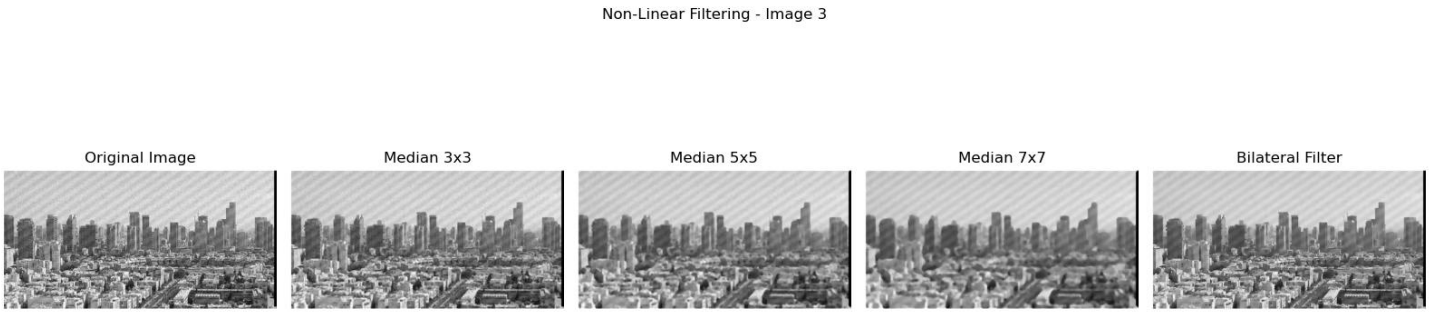


Image 3 processed successfully.

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Periodic/Structured Noise

Observation:

- Median 3×3 provides moderate smoothing while retaining most building details.
- Median 5×5 reduces more fine variations but slightly softens the structures.
- Median 7×7 gives strong smoothing with noticeable loss of fine architectural details.
- Bilateral filtering preserves major building edges better, although some diagonal noise remains.

Conclusion:

- Median 7×7 provides stronger smoothing but causes greater detail loss.
- Bilateral filtering gives better edge preservation with comparatively less blurring.

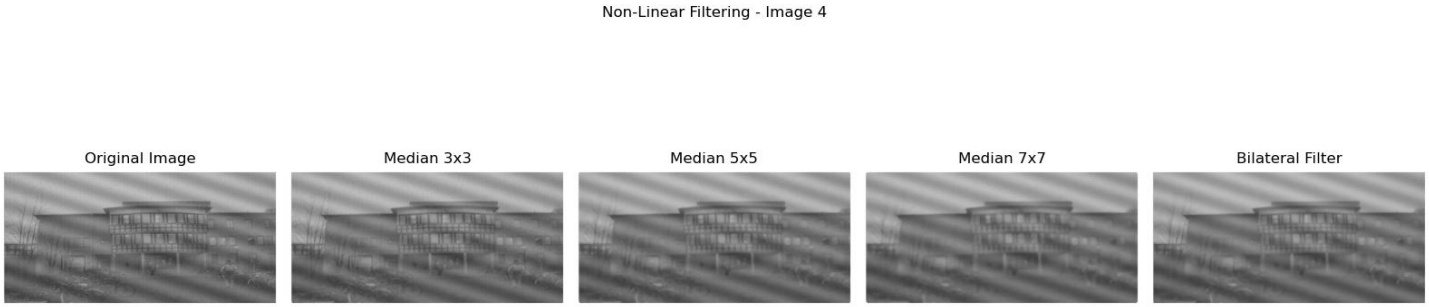


Image 4 processed successfully.
All images processed successfully!

Periodic/Striped Noise

Observation:

- Median 3×3 provides moderate smoothing while maintaining the building structure.
- Median 5×5 gives increased noise reduction with some loss of detail.
- Median 7×7 produces the highest smoothing but makes fine structures softer.

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- ❖ Bilateral filtering preserves the main building structure, but the diagonal noise pattern remains visible

Conclusion:

- ➢ Median filtering gives progressively higher noise reduction as the kernel size increases, but also increases blurring.
- ➢ Bilateral filtering preserves structural details better but does not completely remove the visible noise pattern.

Effect of filter size

Filter	Noise Removal	Detail Preservation	Expected Use
Median 3×3	Moderate	Good	Light impulse noise
Median 5×5	High	Moderate-Good	Best general balance
Median 7×7	Very high	Lower	Strong impulse noise
Bilateral	Moderate	Very good	Edge-sensitive images

Observation

- ✓ The median filtering is most effective for the impulse/salt-and-pepper noise in Images 1 and 2, with Median 5×5 giving a good balance and Median 7×7 providing stronger suppression at the cost of detail.
- ✓ For Images 3 and 4, the visible periodic noise remains after all implemented filters, showing a limitation of these spatial filters.

Practical Uses

- ➲ Salt-and-pepper / impulse noise → Median 3×3 or 5×5
- ➲ Heavy impulse noise → Median 5×5 or 7×7
- ➲ Images requiring strong edge preservation → Bilateral filter
- ➲ Periodic/striped noise → The implemented filters are not optimal; a frequency-domain or periodic-noise-specific method would be more appropriate.

Conclusion

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- The experiment demonstrates that Median 5×5 provides the best overall balance between noise removal and detail preservation for the impulse-noise images.
- Increasing the window to 7×7 improves smoothing but introduces more blurring, while bilateral filtering preserves edges better.
- The periodic noise in Images 3 and 4 remains a failure case, showing that the choice of non-linear filter must depend on the type of noise present.